

Luke Serrano

 Luke Serrano |  lukeserrano0@gmail.com |  (618) 334-5046

EXPERIENCE

Missouri S&T Satellite Research Team (M-SAT)

Aug 2022 - Present

Missouri University of Science & Technology

Rolla, MO

- Chief Engineer of MR/MRS SAT mission
- Building interprocess communication software framework to integrate simulation, GNC, and satellite data handling software libraries
- Managed GitLab repository of GNC algorithm C++ conversion
- Communicate with 10 subsystem leads to ensure proper direction and progression of satellite development
- Troubleshoot SPI, I2C, and USART communication on PCB's to develop interface between microcontrollers and devices

Sandia National Laboratory Internship

Jan 2024 - Present

Guidance, Navigation, and Control

Albuquerque, NM

- Built modular MATLAB script to characterize ACS system to improve fidelity of simulation
- Proving ability to work within industry level software development framework

Northrop Grumman Internship

June 2023 - Aug 2023

Northrop Grumman Space Sector

Gilbert, AZ

- Specialty Systems Engineer Intern at satellite manufacturing facility
- Exposure to industry with requirements, a paying customer, deadlines, and expected mission success
- Performed analyses and created documentation to verify launch requirements from the launch provider

Peer Tutor

Aug 2021 - July 2022

Southwestern Illinois College

Belleville, IL

- Tutored for three semesters at the Success Center, assisting students in mathematics and science
- Collaborated with other peer tutors to address student comprehension challenges
- Improved skill of putting my thoughts into words that can be better understood by others

EDUCATION

Missouri University of Science & Technology

Fall 2022 - Spring 2025

B.S. in Aerospace Engineering

GPA: 3.8

Orbital Mechanics II; Aerodynamics II; Flight Dynamics and Controls I; Spacecraft Design Lab

Southwestern Illinois College

Jun 2021 - Aug 2022

Pursued Associates in Science

GPA: 3.8

SKILLS

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|--------------------------------------|-----------------------|---------------|
| - C++ | - MATLAB | - Hardworking |
| - Collaborative Software Development | - GitLab | - Leadership |
| - Circuitry Board Communication | - Strong Communicator | - Discipline |